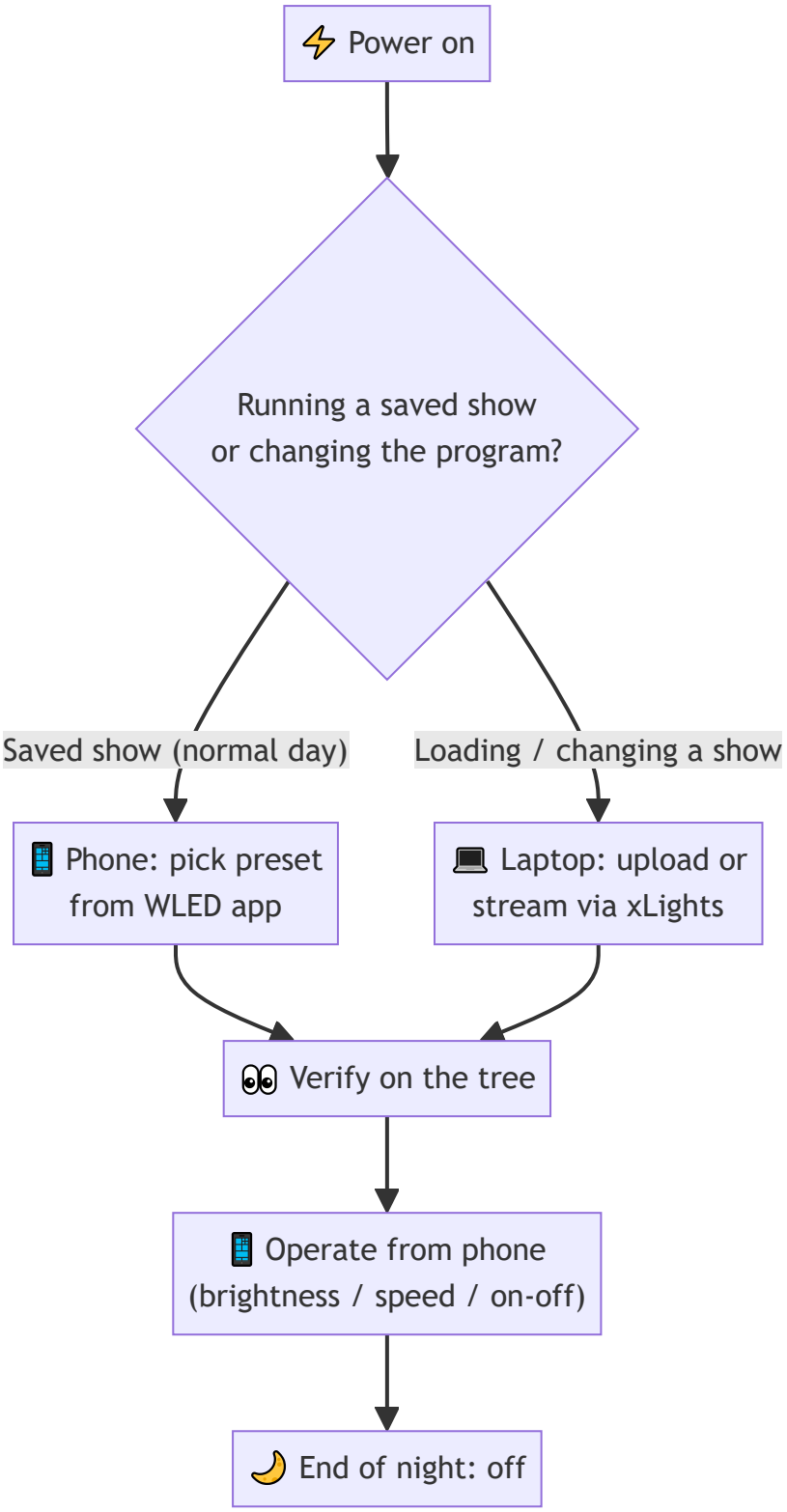
Setup — Day-of Operation

*The runbook you follow **each time you run the tree**. It assumes the controller is already commissioned — the Wi-Fi AP, the xLights 3D model and your saved show library all exist.*

> **First time / changed something?** One-time commissioning (creating the WLED AP, building the xLights 3D cone model, building the show library) lives in the Software Controller Solution → First-time commissioning.

See also: README · Single Page Explanation · Technical Project Document · Software Controller Solution

At a glance



1. Pre-flight checklist

- [ ] Mains is on and both Mean Well PSUs are powered.

- [ ] The controller is broadcasting its AP (`MatrixTree-WiFi` appears in Wi-Fi lists).
- [ ] Phone to hand (and the laptop too, **only** if you're changing the program today).
- [ ] No visible damage to strips, glands or the enclosure seal after weather.

2. Power on

1. Energise the system at the mains (or smart plug, if fitted).
2. Wait for the controller to boot — confirm the `MatrixTree-WiFi` network appears.
3. The tree should light to its last state or default preset.

3. Load the sequence to run today

Choose the path that fits the day:

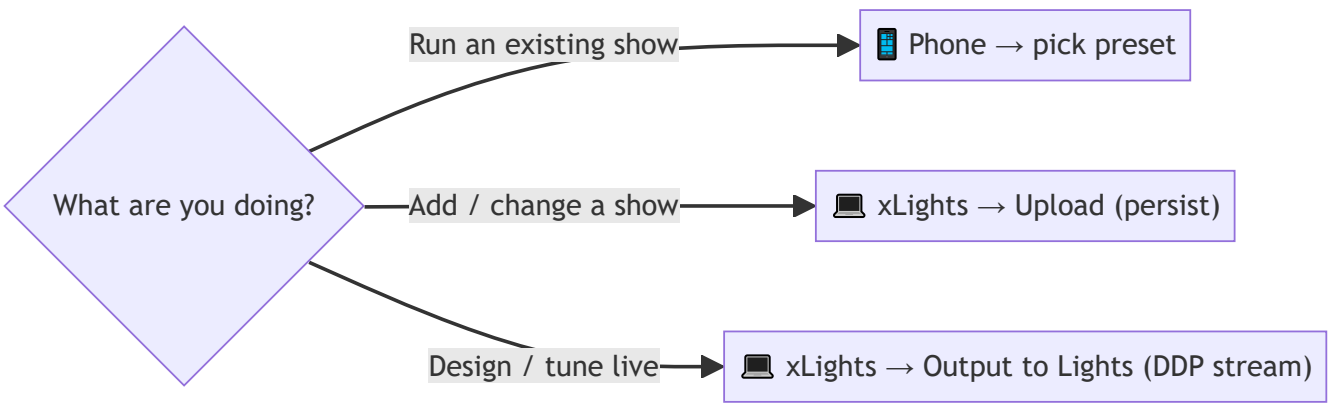
Path A — Phone: select a pre-stored preset *(normal day)*

1. Connect your phone to `MatrixTree-WiFi`.
2. Open the **WLED app** (or browse to the controller's address).
3. Pick the day's **preset / playlist** from the saved library.

> No laptop needed. This is the everyday path.

Path B — Laptop: upload or stream via xLights *(when changing the program)*

1. Connect the laptop to `MatrixTree-WiFi`.
2. Open **xLights**.
3. Then either:
    - **Upload** the chosen sequence to WLED flash so it persists as a new preset (best when you want it available later from the phone), **or**
    - **Output to Lights** (the yellow lightbulb, top-right) to **live-stream** it via DDP for real-time playback/editing.



4. Verify on the tree

- Watch the loaded show start and confirm it maps correctly across the cone.
- If you want a quick mapping sanity check in xLights: right-click the tree model → **Test** → choose a **Bars** or **Butterfly** pattern and look for clean mapping across the real 6 m tree.

5. Operate from the phone

From the WLED app you can, at any time:

- Toggle **on/off** (software master — pixels go black; PSUs stay on).
- Adjust **brightness** and **effect speed**.
- **Pause / resume** or **switch playlist**.

> See the on/off distinction (software blackout vs cutting mains) in the Software Controller Solution → on/off.

6. End of night

- **Software off:** tap the master power toggle in the WLED app — quickest, leaves the system on standby.
- **True power-down:** cut mains (or use the smart plug) when you want zero standby draw or before maintenance.

Quick reference card

Setting	Value
Wi-Fi network (AP)	`MatrixTree-WiFi`
AP password	*(your custom password, set at commissioning)*
Controller dashboard	`4.3.2.1`
xLights output protocol	DDP
xLights controller IP	`4.3.2.1`
Strings × nodes	12 × 360 = 4,320 pixels
LED1 — GPIO 16	start 0, count 1,080
LED2 — GPIO 3	start 1,080, count 1,080
LED3 — GPIO 1	start 2,160, count 1,080
LED4 — GPIO 4	start 3,240, count 1,080
Default WLED network (factory)	`WLED-AP` / `wled1234`